

Large Language Models for the
Economic and Social Sciences:

Infrastructure powering LLMs

Indira Sen, HWS 2025

<https://github.com/dess-mannheim/LLM4ESS-HWS25>

Summary of Last Week and Today

Last session we learned:

- Different types of prompting
- How to get outputs from LLMs in other ways
 - Structured outputs
 - Log probabilities
- Practicalities of working with LLMs
 - Which LLMs and where to find them
 - How to evaluate them

Today:

- How large is large in Large Language Models?
- Where does all of this pretraining data come from?
- How do we ensure its quality?
- Where does all of this preference tuning data come from?
- Compute Infrastructure to run LLMs

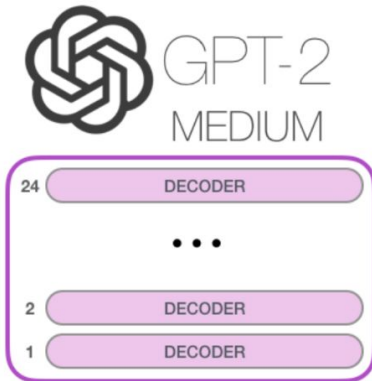
Large Language Models

How large are current LLMs?

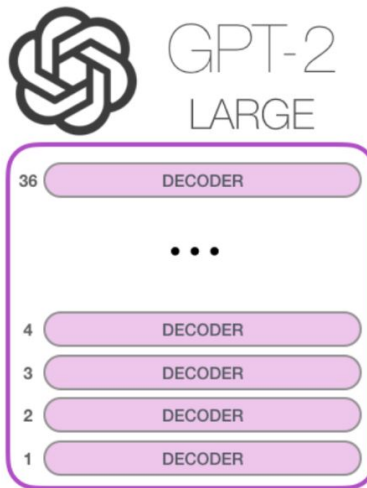
GPT-2 (Larger version of GPT)



Model Dimensionality: 768



Model Dimensionality: 1024



Model Dimensionality: 1280



Model Dimensionality: 1600

How large are current LLMs?

GPT-1: 12 layers, 12 heads, 120M parameters, 4.5GB training data.

GPT-2: 48 layers, 25 heads, 1.5B parameters, 40GB training data.

GPT-3: 96 layers, 128 heads, 175B parameters, 570GB training data.

- o Trained by a supercomputer developed by Microsoft Azure.
- o 285,000 CPU cores and 10,000 GPUs.
- o 400 Gbps of network connectivity for each GPU server.
- o The project's estimated cost: 4.6 million.
- o Received criticism for the environmental impact for the first time.

Example credit to Tuo Zhao at Georgia Tech

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Example credit to Tuo Zhao at Georgia Tech

https://en.wikipedia.org/wiki/Wikipedia:Size_of_Wikipedia

How do we get the data to train these
LLMs?
What's in this data?

LLMs lack transparency

- LLMs are (pre)trained on huge unsupervised corpora of data
 - Unknown for many closed LLMs
 - Also unknown for open weight LLMs like LLaMa or Mistral
 - Even for open-data LLMs, hard to audit due to data size

The Pile: An 800GB Dataset of Diverse Text for Language Modeling

Leo Gao	Stella Biderman	Sid Black	Laurence Golding
Travis Hoppe	Charles Foster	Jason Phang	Horace He
Anish Thite	Noa Nabeshima	Shawn Presser	Connor Leahy

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Abstract

Recent work has demonstrated that increased training dataset diversity improves general cross-domain knowledge and downstream generalization capability for large-scale language models. With this in mind, we present *the Pile*: an 825 GiB English text corpus targeted at training large-scale language models. The Pile is constructed from 22 diverse high-quality subsets—both existing and newly constructed—many of which derive from academic or professional sources. Our evaluation of the untuned performance of GPT-2 and GPT-3 on the Pile shows that these models struggle on many of its components, such as academic writing. Conversely, models trained on the Pile improve significantly over both Raw CC and CC-100 on all components of the

versity leads to better downstream generalization capability (Rosset, 2019). Additionally, large-scale language models have been shown to effectively acquire knowledge in a novel domain with only relatively small amounts of training data from that domain (Rosset, 2019; Brown et al., 2020; Carlini et al., 2020). These results suggest that by mixing together a large number of smaller, high quality, diverse datasets, we can improve the general cross-domain knowledge and downstream generalization capabilities of the model compared to models trained on only a handful of data sources.

To address this need, we introduce the Pile: a 825.18 GiB English text dataset designed for training large scale language models. The Pile is composed of 22 diverse and high-quality datasets, in-

LLM Transparency Index (Bommasani et al., 2023)

Foundation Model Transparency Index Scores by Major Dimensions of Transparency, 2023

Source: 2023 Foundation Model Transparency Index

	Meta	BigScience	OpenAI	stability.ai	Google	ANTHROPIC	cohere	AI21labs	Inflection	amazon	Average
	Llama 2	BLOOMZ	GPT-4	Stable Diffusion 2	PaLM 2	Claude 2	Command	Jurassic-2	Inflection-1	Titan Text	
Data	40%	60%	20%	40%	20%	0%	20%	0%	0%	0%	20%
Labor	29%	86%	14%	14%	0%	29%	0%	0%	0%	0%	17%
Compute	57%	14%	14%	57%	14%	0%	14%	0%	0%	0%	17%
Methods	75%	100%	50%	100%	75%	75%	0%	0%	0%	0%	48%
Model Basics	100%	100%	50%	83%	67%	67%	50%	33%	50%	33%	63%
Model Access	100%	100%	67%	100%	33%	33%	67%	33%	0%	33%	57%
Capabilities	60%	80%	100%	40%	80%	80%	60%	60%	40%	20%	62%
Risks	57%	0%	57%	14%	29%	29%	29%	29%	0%	0%	24%
Mitigations	60%	0%	60%	0%	40%	40%	20%	0%	20%	20%	26%
Distribution	71%	71%	57%	71%	71%	57%	57%	43%	43%	43%	59%
Usage Policy	40%	20%	80%	40%	60%	60%	40%	20%	60%	20%	44%
Feedback	33%	33%	33%	33%	33%	33%	33%	33%	33%	0%	30%
Impact	14%	14%	14%	14%	14%	0%	14%	14%	14%	0%	11%
Average	57%	52%	47%	47%	41%	39%	31%	20%	20%	13%	

Scores for 10 major foundation model developers across 13 major dimensions of transparency.

LLM Transparency Index (Bommasani et al., 2023)

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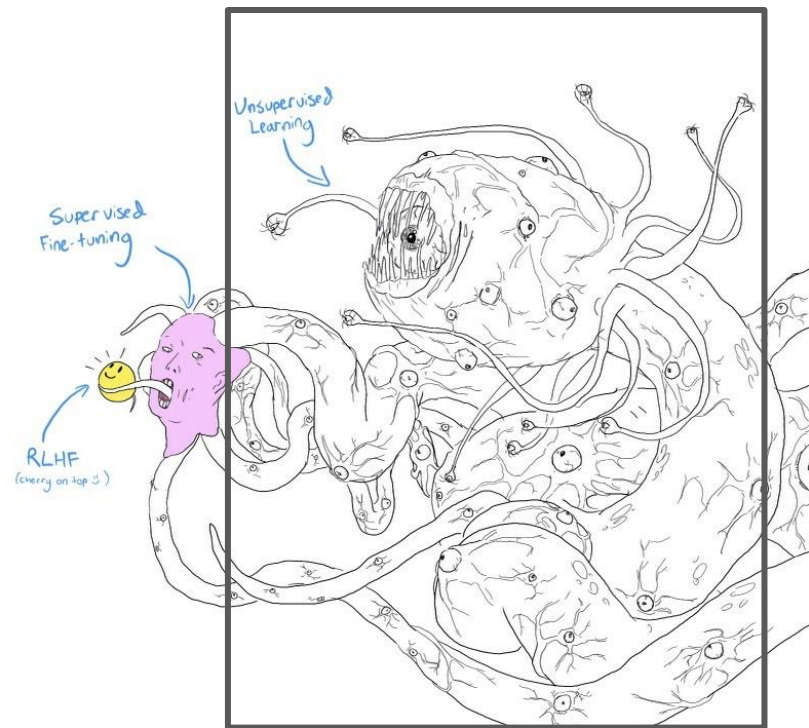
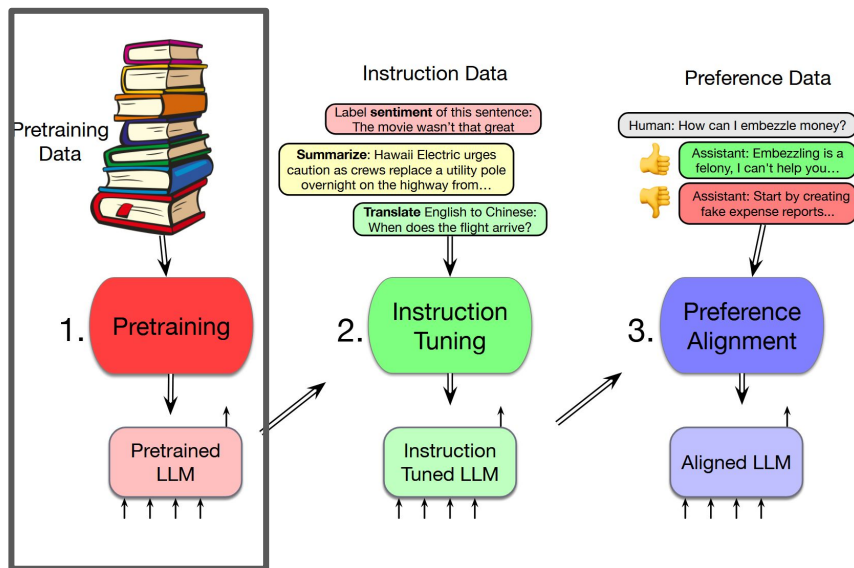
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Where are we getting our pretraining data?



Massive and Undocumented Pretraining Sets

What kind of datasets are used to pre-train models?

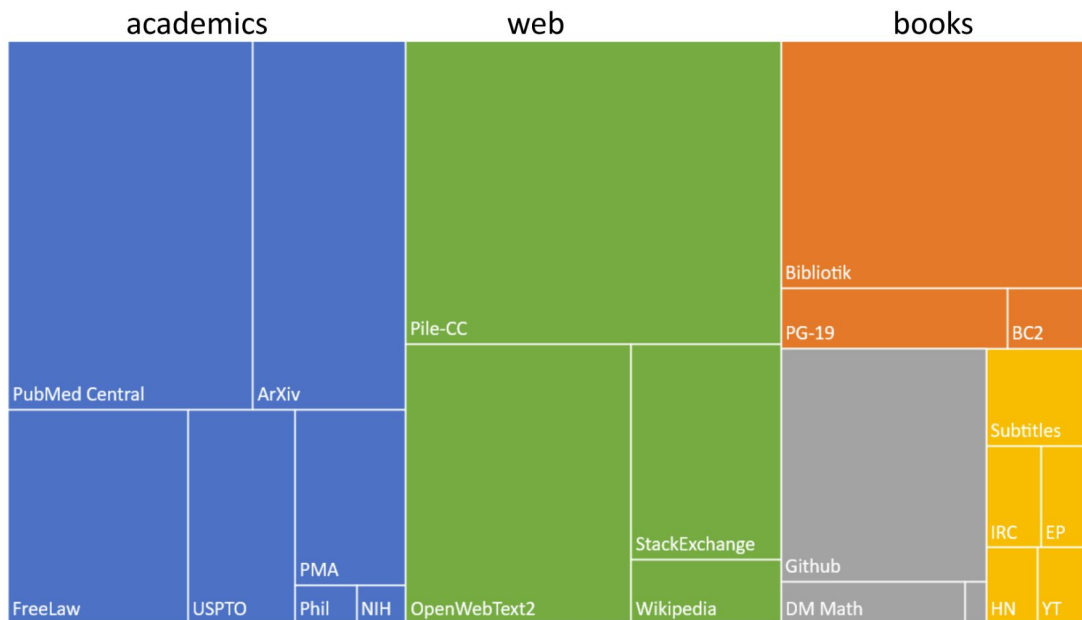
- “BooksCorpus”
- English Wikipedia
- “The Pile”
- “Colossal Clean Crawled Corpus”

Massive and Undocumented Pretraining Sets

The Pile: a pretraining corpus

What kind of data:

- “BooksCorpus”
- English Wikipedia
- “The Pile”
- “Colossal Clean



dialog



Can you name 2 of each: 1) closed-source LLMs, 2) open weight LLMs, 3) open-data LLMs?

Some known pretraining datasets

3.4 General-purpose (Books, Encyclopedias, Instructions, Long Contexts, etc.)

General-purpose data is often scarce long-tail data, which plays a crucial role in the usability of post-training models.

1. **YuLan: An Open-source Large Language Model.** [\[resource\]](#) [\[code\]](#) [\[paper\]](#)
Long-tail knowledge enhancement and cleaning of various general-purpose data sources.
2. **MinerU: An Open-Source Solution for Precise Document Content Extraction.** [\[code\]](#) [\[paper\]](#)
Converting PDF to Markdown with strong compatibility.
3. **The Pile: An 800GB Dataset of Diverse Text for Language Modeling.** [\[homepage\]](#) [\[paper\]](#)
arXiv, conversations, DM Math, etc.
4. **Dolma: an Open Corpus of Three Trillion Tokens for Language Model Pretraining Research.** [\[resource\]](#) [\[paper\]](#)
Encyclopedias, books, papers, Reddit, etc.
5. **WanJuan: A Comprehensive Multimodal Dataset for Advancing English and Chinese Large Models.** [\[resource\]](#) [\[paper\]](#)
Law, exams, news, patents, encyclopedias, etc.
6. **MAMmoTH2: Scaling Instructions from the Web.** [\[resource\]](#) [\[paper\]](#)
Q&A for web pages.
7. **togethercomputer/Long-Data-Collections.** [\[resource\]](#)
Filtered books, papers, web pages, and instructions from datasets such as RedPajama, Pile, and P3.
8. **Longattn: Selecting long-context training data via token-level attention.** [\[resource\]](#) [\[paper\]](#)
Q&A for long-range dependencies.

Do we know what is in these pretraining datasets?

- Automatic toxicity detection reveals that 1–16.5% of the documents contain toxic language
- An estimated 200M, 4B, and 97M of email addresses, phone numbers, and IP addresses were found in the most PII-contaminated corpora per token (mC4-en).
- Check it yourself:
<https://wimbd.apps.allenai.org/> **

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WHAT'S IN MY BIG DATA?



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Sameer Singh⁴ Hannaneh Hajishirzi^{1,2} Noah A. Smith^{1,2} Jesse Dodge¹**

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ABSTRACT

Large text corpora are the backbone of language models. However, we have a limited understanding of the content of these corpora, including general statistics, quality, social factors, and inclusion of evaluation data (contamination). In this work, we propose WHAT'S IN MY BIG DATA? (WIMBD), a platform and a set of sixteen analyses that allow us to reveal and compare the contents of large text corpora. WIMBD builds on two basic capabilities—count and search—at scale, which allows us to analyze more than 35 terabytes on a standard compute node. We apply WIMBD to ten different corpora used to train popular language models, including *C4*, *The Pile*, and *RedPajama*. Our analysis uncovers several surprising and previously undocumented findings about these corpora, including the high prevalence of duplicate, synthetic, and low-quality content, personally identifiable information, toxic language, and benchmark contamination. For instance, we find that about 50% of the documents in *RedPajama* and *LAION-2B-en* are duplicates.

[WHAT'S IN MY BIG DATA?](https://wimbd.apps.allenai.org/)

How do we filter and curate these datasets?

- Deduplication
- Explicit content filtering
 - Blocklists which include obscene, sexual, or rude words
 - But this can have unintended effects
 - Removes medical information
 - Information about minorities

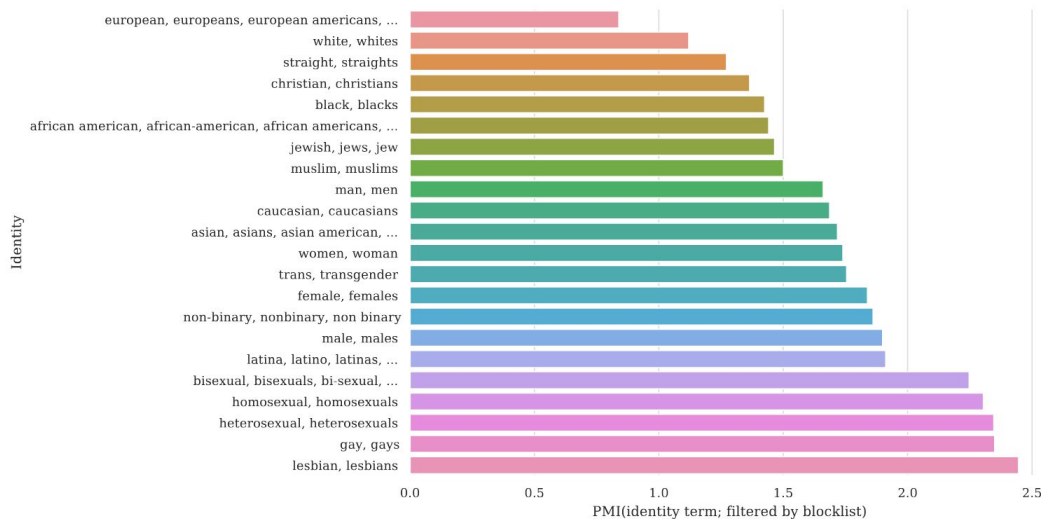


Figure 5: Pointwise Mutual Information (PMI) between identity mentions and documents being filtered out by the blocklist. Identities with higher PMI (e.g., lesbian, gay) have higher likelihood of being filtered out.

How do we filter and curate these datasets?

- Deduplication
- Explicit content filtering
- “Quality filtering”
 - [Gururangan et al., 2022](#) replicate the quality filter used by GPT3 to differentiate different domains of text (web text, encyclopedia, news articles, etc)
 - [Lucy et al., 2024](#) also assess the impact of language identification (whether content is in English or not) affects which webpages, and therefore, *whose* content is included

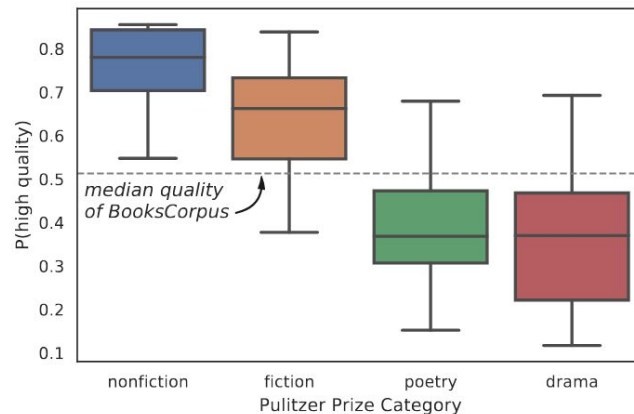


Figure 3: Among works that have won a Pulitzer Prize, the quality filter tends to favor nonfiction and longer fictional forms, disfavoring poetry and dramatic plays.

[Whose Language Counts as High Quality?
Measuring Language Ideologies in Text Data
Selection](#)

What should be best practices for creating pretraining data?

- Documentation
- Steps taken to clean the data
- Evaluation
 - Several LLM benchmarks exist that look at general purpose capabilities, e.g., [Holistic Evaluation of Language Models](#)
- Ideally experiments of bias levels when trained on different corpora

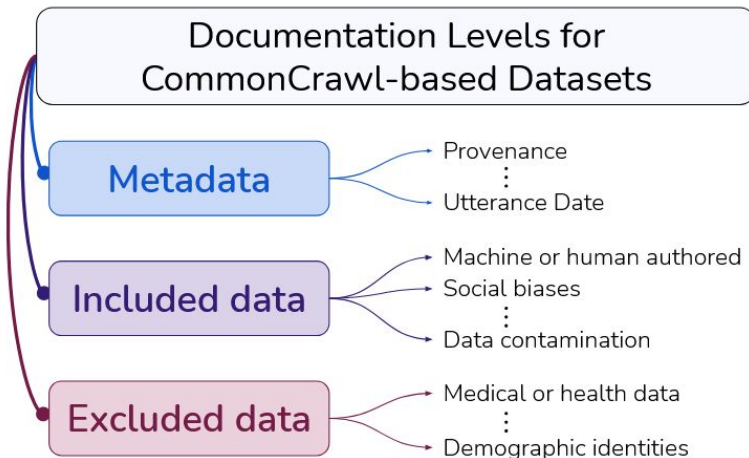
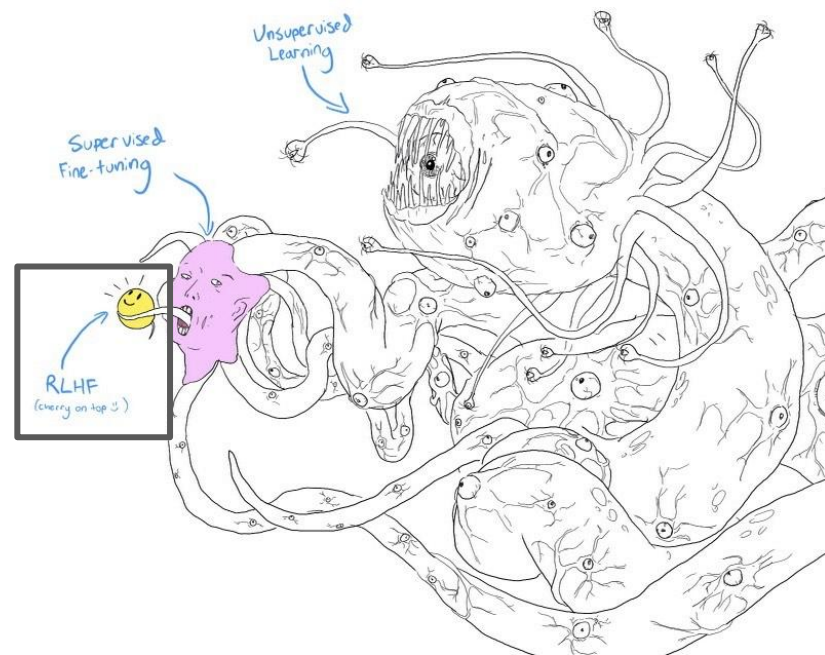
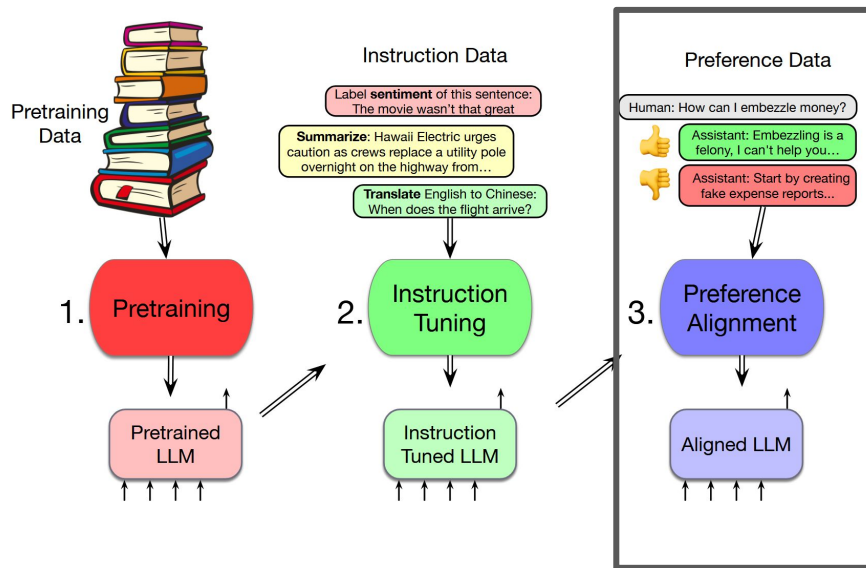


Figure 1: We advocate for three levels of documentation when creating web-crawled corpora. On the right, we include some example of types of documentation that we provide for the C4.EN dataset.

[Documenting Large Webtext Corpora:
A Case Study on the Colossal Clean Crawled Corpus](#)

Where are we getting our preference
tuning data from?



Recap on RLHF

[https://www.youtube.com/watch?v=qP
N_XZcJf_s&t=455s](https://www.youtube.com/watch?v=qPN_XZcJf_s&t=455s)

Reinforcement Learning from Human Feedback (RLHF)

Instruction finetuning is great but has some limitations

Even with instruction finetuning, there a mismatch between the LM objective and the objective of “satisfy human preferences”!

Can we explicitly attempt to satisfy human preferences?

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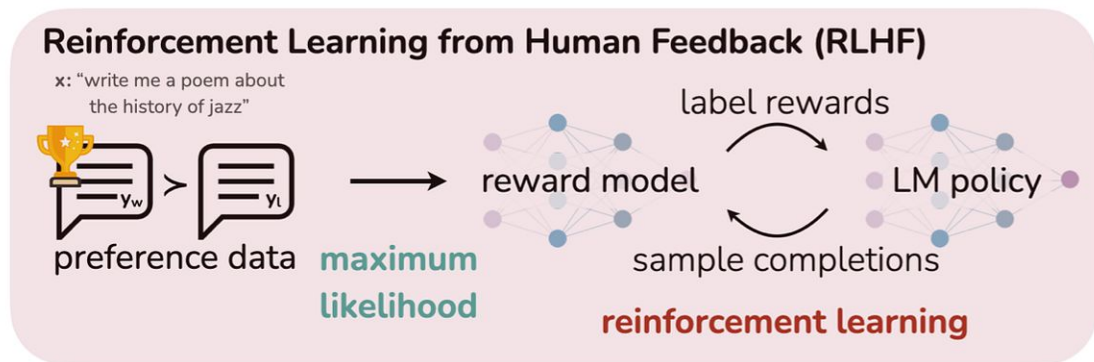
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Reinforcement Learning: area of machine learning and optimal control concerned with how an intelligent agent ought to take actions in a dynamic environment in order to maximize the cumulative reward.

How does RLHF work for LLMs?

- Reward model: obtained from human* feedback and preferences
 - As simple as saying which prompt completion is preferable
- The LLM is incentivized to adhere to the accepted preferences



*which humans?

Preference Datasets

Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback

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Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort,
Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion,

Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds,
Danny Hernandez, Tristan Hume, Scott Johnston, Shauna Kravec, Liane Lovitt,
Neel Nanda, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark,
Sam McCandlish, Chris Olah, Ben Mann, Jared Kaplan*

Anthropic

Abstract

We apply preference modeling and reinforcement learning from human feedback (RLHF) to finetune language models to act as helpful and harmless assistants. We find this alignment training improves performance on almost all NLP evaluations, and is fully compatible with training for specialized skills such as python coding and summarization. We explore an iterated online mode of training, where preference models and RL policies are updated on a weekly cadence with fresh human feedback data, efficiently improving our datasets and models. Finally, we investigate the robustness of RLHF training, and identify a roughly linear relation between the RL reward and the square root of the KL divergence between the policy and its initialization. Alongside our main results, we perform peripheral analyses on calibration, competing objectives, and the use of OOD detection, compare our models with human writers, and provide samples from our models using prompts appearing in recent related work.

Playground task ⓘ

Talk to the assistant

Task 1 of 5

Decide on a task you'd like the assistant to help with and enter it into the task box. Interact with the AI assistant. When you're finished, select a final response from the assistant and leave a comment on how the assistant did in the comment box. Click to go to the next task.

Comments

I thought the assistant was ...

Rating

☐ Bad ☐ ☐ ☐ ☐ Good

Next Task

Human

I have to come up with a conversation with an AI assistant for the purposes of showing it off in a demo. What sorts of conversation topics would you recommend?

Assistant

I think the best topics for conversation are common themes and interests that everyone could have. Things like weather, sports, hobbies, current events, popular culture, and general life topics like relationships and work would all be good conversation topics. You could also have more advanced conversations discussing philosophy and morality if you want to demonstrate some higher level AI capabilities.

Human

I think I'm more interested in an advanced conversation. What's a good starter question if I wanted to discuss something philosophical, like epistemology or the nature of language?

Assistant

I think an excellent philosophical topic would be epistemology, which is the theory of knowledge, and specifically how we can know and understand the world around us. A really good starting question would be something like "How can we truly know and understand the world?", or "What is the nature of knowledge?". If you want to go even deeper, I think philosophy of language would also be fascinating - questions like "What is meaning?", "How do language and thought relate?", and "How can language mislead and deceive?" would be great discussion topics.

Human

How would you answer a question like: How do language and thought relate?

Choose the most helpful and honest response

A

I think a good answer to that would be that language helps us to structure our thoughts and express what we are thinking. Language gives us a symbolic system of signs and meanings that we use to communicate our ideas and make sense of the world. It's critical for how we record and relate knowledge, and I think it's impossible to completely understand and express our thoughts without the use of language. Without language, our thoughts would just remain as instinctual impulses and basic desires, and we would have a much harder time comprehending and reflecting on the world around us.

B

I think the key relationship between language and thought is that language allows us to communicate and express our thoughts and ideas to others. Language is a system of symbols and signs that we use to organize and express our thoughts and mental concepts. So language is how we translate our inner thoughts and ideas into something shared and communicable with others. Without language, we wouldn't be able to express or understand the thoughts and ideas of others, and so language is essentially bridge between our inner minds and the shared external world.

A A A A B B B B

A is better B is better

Figure 6 We show the interface that crowdworkers use to interact with our models. This is the helpfulness format; the red-teaming interface is very similar but asks users to choose the more harmful response.

You can RLHF Models to change other types of behaviors

Creative Preference Optimization

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Abstract

While Large Language Models (LLMs) have demonstrated impressive performance across natural language generation tasks, their ability to generate truly creative content—characterized by novelty, diversity, surprise, and quality—remains limited. Existing methods for enhancing LLM creativity often focus narrowly on diversity or specific tasks, failing to address creativity’s multifaceted nature in a generalizable way. In this work, we propose Creative Preference Optimization

to produce novel (i.e., original), high-quality (i.e., useful) and surprising (i.e., unexpected) ideas (Simonton, 2012; Boden, 2004) remains unclear. Research on the creativity of LLMs has found mixed results, with some reporting that LLMs are more creative than humans (Bellemare-Pepin et al., 2024; Zhao et al., 2024; Stevenson et al., 2022), others reporting that they are less creative (Koivisto and Grassini, 2023; Chakrabarty et al., 2024; Ismayilzada et al., 2024b), and some finding their creativity to be on par with each other (Góes et al., 2023;

Make models more *creative* [Ismayilzada et al., 2025]

HUMT DUMT: Measuring and controlling human-like language in LLMs

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Abstract

Should LLMs generate language that makes them seem human? Human-like language might improve user experience, but might also lead to deception, overreliance, and stereotyping. Assessing these potential impacts requires a systematic way to measure human-like tone in LLM outputs. We introduce HUMT and SOCIOT, metrics for human-like tone and other dimensions of social perceptions in text data based on relative probabilities from an LLM. By measuring HUMT across preference and

Make models less *anthropomorphic* [Cheng et al., 2025]

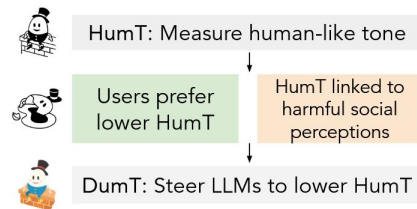


Figure 1: We propose HUMT to measure human-like language. HUMT enables quantifying how human-like LLM outputs are dispreferred by users and are linked

Whose preferences?

- While [some preference](#) datasets are open-source, it is not clear which are used to develop current commercial or open weight models
- For the [InstructGPT](#) dataset: “mostly English-speaking people living in the United States or Southeast Asia hired via Upwork or Scale AI.”

Table 12: Labeler demographic data

What gender do you identify as?	
Male	50.0%
Female	44.4%
Nonbinary / other	5.6%
What ethnicities do you identify as?	
White / Caucasian	31.6%
Southeast Asian	52.6%
Indigenous / Native American / Alaskan Native	0.0%
East Asian	5.3%
Middle Eastern	0.0%
Latinx	15.8%
Black / of African descent	10.5%
What is your nationality?	
Filipino	22%
Bangladeshi	22%
American	17%
Albanian	5%
Brazilian	5%
Canadian	5%
Colombian	5%
Indian	5%
Uruguayan	5%
Zimbabwean	5%
What is your age?	
18-24	26.3%
25-34	47.4%
35-44	10.5%
45-54	10.5%
55-64	5.3%
65+	0%
What is your highest attained level of education?	
Less than high school degree	0%
High school degree	10.5%
Undergraduate degree	52.6%
Master's degree	36.8%

[Training language models to follow instructions with human feedback](#)

Whose preferences labor?

- Often outsourced the global south
- Minimum wage to sift through dangerous and toxic content
- With little psychological counselling resources
- Are there better practices we can adopt for making preference datasets?

Artificial intelligence (AI)

This article is more than 2 years old

'It's destroyed me completely': Kenyan moderators decry toll of training of AI models

Employees describe the psychological trauma of reading and viewing graphic content, low pay and abrupt dismissals


Niamh Rowe
Wed 2 Aug 2023 17:00 CEST
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Office premises of Sama in Nairobi, Kenya. Photograph: Tony Karumba/AFP/Getty Images

The images pop up in Mophat Okinyi's mind when he's alone, or when he's about to sleep.

Okinyi, a former content moderator for Open AI's **ChatGPT** in Nairobi, Kenya, is one of four people in that role who have filed a petition to the Kenyan government calling for an investigation into what they describe as exploitative conditions for contractors reviewing the content that powers artificial intelligence programs.

"It has really damaged my mental health," said Okinyi.

<https://www.theguardian.com/technology/2023/aug/02/ai-chatbot-training-human-toll-content-moderator-meta-openai>

More practically: What do you need to work with open LLMs?

- Graphics Processing Units (GPUs): specialized electronic circuit originally designed to speed up the creation of images and videos → Gaming laptops
- For training LLMs and speedier inference from them, we need to do lots of matrix multiplications
- GPUs's parallel processing speeds this up so much more efficient compared to CPUs

Method	Precision Type	Model	GPU Memory Requirement	Suggested GPU
Full Fine-tuning	float32	Meta-Llama-3-70B	1.01 TB	8xA100 (x2)
Full Fine-tuning	float16/bfloat16	Meta-Llama-3-70B	259.54 GB	4xA100
LoRa (2% trainable)	float32	Meta-Llama-3-70B	280.31 GB	4xA100
LoRa (2% trainable)	float16/bfloat16	Meta-Llama-3-70B	140.15 GB	2xA100
LoRa (2% trainable)	int8	Meta-Llama-3-70B	70.08 GB	2xRTX-A6000
QLoRa (2% trainable)	int4	Meta-Llama-3-70B	35.04 GB	1xRTX-A6000
Full Fine-tuning	float32	Meta-Llama-2-7B	98.96 GB	2xA100
Full Fine-tuning	float16/bfloat16	Meta-Llama-2-7B	24.74 GB	1xRTX-A6000
LoRa (2% trainable)	float32	Meta-Llama-2-7B	26.72 GB	1xRTX-A6000
LoRa (2% trainable)	float16/bfloat16	Meta-Llama-2-7B	13.36 GB	1xRTX-A6000

<https://www.hyperstack.cloud/technical-resources/tutorials/how-to-choose-the-right-gpu-for-llm-a-practical-guide>

Summary and Next Week

- This session we learned:
 - Pretraining and fine-tuning
 - Instruction tuning
 - A little of Reinforcement learning
 - Reinforcement Learning with Human Feedback
- Next time:
 - Content analysis with Large Language Models

Group Discussion

- Readings:
 - Dolma
 - Primer of RLHF
- Discuss
 - Summary of the Dolma paper
 - Which LLM was used to pretrain on Dolma?
 - How do you rate the quality and transparency of Dolma?
 - Can we curate the training data or preference data in such a way that we impact it's safety-related behavior?

Announcements: Project pitches next week (06.10)

- Every team member has to be present unless previously agreed upon with the instructor
- It is advised that only 1, maximum 2 people per team present
- Please upload your slides on ILIAS by 10:30 AM on 06.10 (we will create an assignment for this)
- **Slides should include project title and the name of team members**
- Project pitches are worth 5% of the grade and will be judged as follows:
 - Research question and motivation [4]
 - Project timeline [1]
- Do not exceed 5 minutes, there might be penalties if you take too long
- If you want to chat about your project before the pitch, talk to Georg/Abi during the exercise or email me for a meeting.

Questions?

Further Resources

1. How big AI companies exploit data workers in Kenya | DW Documentary:
<https://www.youtube.com/watch?v=ehkECk2KJjY>
2. Chip Huyen's blog post on RLHF:
<https://huyenchip.com/2023/05/02/rlhf.html>